curriculum vitæ of
Chuan Guo
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Research Interests

Fields: Computer Vision, Computer Graphics, Machine Learning
Topics: Generative AI for Digital Humans and Character Animation, Human-world Interactions.

Employment

Aug 2025 – Present | Reality labs, Meta Inc. | Redmond, USA
Senior Research Scientist

May 2024 – Aug 2025 | Snap Research, Snap Inc. | New York, USA
Research Scientist

Sep. 2019 – Jan. 2024 | Vision and Learning Lab, University of Alberta | Edmonton, Canada
Research Assistant. Supervisor: Prof. Li Cheng

Nov. 2022 – Dec. 2023 | Huawei Technologies Canada Co., Ltd. | Markham, Canada
Associate Researcher, Intern. Mentor: Dr. Juwei Lu

Jan. 2021 – Jan. 2022 | Huawei Technologies Canada Co., Ltd. | Edmonton, Canada
Associate Researcher, Intern. Mentor: Wei Lu

Education

Sep. 2019 – Jan. 2024 | **Ph.D.** in Software Engineering and Intelligent Systems | University of Alberta, Canada
Dept. of Electrical and Computer Engineering
- Advisor: Prof. Li Cheng
- Thesis: Deep Learning for 3D Human Action Modeling and Understanding

Sep. 2013 – Jul. 2017 | **B.Eng.** in Software Engineering (Pilot Program) | Jilin University, China
College of Software

Projects

Aug 2025 – Present | **Social Embodied Intelligence and Human-world Interactions** | Meta Reality Labs

1. Develop models for autonomous user-agent dyadic conversational motion with 3D scene-aware interactions.
2. Investigate ego-vision understanding and scene voxelization as complementary representations for human-world interaction modeling.
3. Contribute to motion foundation model development, post-training, and downstream adaptation.

May 2024 – Aug 2025 | **Generative AI for 3D Character Animation** | Snap Research

1. Design, implement, and experiment AI models that enable automated production of 3D animation of expressive styles and precise control [16].
2. Collect and annotate high-quality large-scale 3D motion capture data with expressive texts [13].
3. Lead research projects on 3D human interaction synthesis [5, 7, 6] and re-targeting [8].

Jan. 2021 – Dec. 2023 | **Language Grounded 3D Human Behavior Modeling and Understanding** | University of Alberta

1. Aimed to synthesize 3d human behaviors from text descriptions or in the inverse way, i.e. understand human behaviors through texts.
2. Annotated so-far the largest motion-language dataset, with 15k motions captioned by 50k descriptions.
3. A novel approach that generates realistic human motion with temporal VAE and RNNs (Demo) [24].
4. Built mutual mappings between 3D human motions and texts, motion captioning and text2motion generation respectively, using vector quantization and Transformers. [23]
5. A motion generator and editor based on generative masked Transformer and residual quantization. [18].

Feb. 2023 – Dec. 2023 | **Generative Human Motion Stylization** | Huawei Technologies Canada

1. The goal is to stylize an existing 3D motion with style clues from for example, motion, label or priors.
2. Found that stylizing motion in latent space is more efficient than in pose space.
3. Designed a generative framework which enables diverse and novel stylization (Demo) [20].

Sep. 2019 – Jan. 2021 **3D Human Action and Video Generation** University of Alberta

1. The topic is to generate human behaviors conditioned on action categories.
2. Synthesized visually pleasing human motions by a novel VAE-based (i.e. Variational AutoEncoder) network with Lie pose representation, and curated own dataset (Demo Webpage) [25].
3. Built up a novel pipeline to generate human videos from action type & single image with graphics & machine learning apparatus (Demo) [19].

† Corresponding author.

Publications

† *Corresponding author.*

Topic: Human-Human/Scene/Object Interaction

[1] Zhang, Zimu, Yucheng Zhang, Xiyan Xu, Ziyin Wang, Sirui Xu, Kai Zhou, Bing Zhou, **Chuan Guo**, Jian Wang, Yu-Xiong Wang†, Liang-Yan Gui†. "*HandX: Scaling Bimanual Motion and Interaction Generation.*" In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2026. (Accept rate: 25.4%)

[2] Wu, Qingxuan, Zhiyang Dou, Chuan Guo, Yiming Huang, Qiao Feng, Bing Zhou, Jian Wang, and Lingjie Liu. "*Text2interact: High-fidelity and diverse text-to-two-person interaction generation.*" In International Conference on Learning Representations (ICLR). 2026. (Accept rate: 27.4%)

[3] Wang, Ziyin, Sirui Xu, **Chuan Guo**, Bing Zhou, Jiangshan Gong, Jian Wang, Yu-Xiong Wang†, and Liang-Yan Gui†. "*Unleashing Guidance Without Classifiers for Human-Object Interaction Animation.*" In International Conference on Learning Representations (ICLR). 2026. (Accept rate: 27.4%)

[4] Sui, Keiwei, Anindita Ghosh, Inwoo Hwang, Jian Wang, **Chuan Guo**†. "*A Survey on Human Interaction Motion Generation*." International Journal of Computer Vision (IJCV). 2025.

[5] Hwang, Inwoo, Bing Zhou†, Young Min Kim, Jian Wang, **Chuan Guo**†. "*SceneMI: Motion In-betweening for Modeling Human-Scene Interactions*." IEEE International Conference on Computer Vision (ICCV, Highlight). 2025. (Accept rate: 24%)

[6] Ghosh, Anindita, Bing Zhou†, Rishabh Dabral, Jian Wang, Vladislav Golyanik, Christian Theobalt, Philipp Slusallek, **Chuan Guo**†. "*DuetGen: Music Driven Two-person Dance Generation via Hierarchical Masked Modeling*." In Proceeding of ACM SIGGRAPH. 2025.

[7] Liu, Shaowei, **Chuan Guo**†, Bing Zhou†, Jian Wang†."*Ponimator: Unfolding Interactive Pose for Versatile Human-human Interaction Animation*." IEEE International Conference on Computer Vision (ICCV). 2025. (Accept rate: 24%)

[8] Wan, Weilin, **Chuan Guo**†, Jian Wang, Bing Zhou†."*InterRET: Self-Supervised Multi-Character Motion Retargeting*." Pre-print (Under review). 2025.

[9] Gohar, Muhammad, **Chuan Guo**, Li Cheng†, Xingyu Li. "*InterMask: 3D Human Motion Interaction Generation via Collaborative Masked Modeling*." In International Conference on Learning Representations (ICLR). 2025. (Accept rate: 32%)

[10] Wang, Yilin, **Chuan Guo**, Li Cheng†, Hai Jiang "*RegionGrasp: A Novel Task for Contact Region Controllable Hand Grasp Generation*." European Conference on Computer Vision Workshop (ECCVW). 2024.

Topic: Motion Foundational Models

[11] Li, Zekun, Sizhe An, Chengcheng Tang, **Chuan Guo**, Ivan Shugurov, Linguang Zhang, Amy Zhao, Srinath Sridhar, Lingling Tao, and Abhay Mittal†. "*LLaMo: Scaling Pretrained Language Models for Unified Motion Understanding and Generation with Continuous Autoregressive Tokens.*" In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2026. (Accept rate: 25.4%)

[12] Cong, Xiaoyan, Zekun Li, Zhiyang Dou, Hongyu Li, Omid Taheri, **Chuan Guo**, Abhay Mittal et al. "*UMO: Unified In-Context Learning Unlocks Motion Foundation Model Priors.*" arXiv preprint arXiv:2603.15975 (2026).

[13] **Guo, Chuan**†, Inwoo Hwang, Jian Wang, Bing Zhou†. "*SnapMoGen: Human Motion Generation from Expressive Texts*." The Thirty-Ninth Annual Conference on Neural Information Processing Systems. 2025. (Accept rate: 24.5%)

Topic: Human Motion Synthesis and Manipulation

[14] Li, Zhengxuan, Qinhui Yang, Yiyu Zhuang, **Chuan Guo**, Xinxin Zuo, Xiaoxiao Long, Yao Yao, Xun Cao, Qiu Shen, and Hao Zhu†. "*Pressure2Motion: Hierarchical Human Motion Reconstruction from Ground Pressure with Text Guidance.*" In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2026. (Accept rate: 25.4%)

[15] Zhong, Lei, **Chuan Guo**, Yiming Xie, Jiawei Wang, Changjian Li†. "*Sketch2Anim: Transferring Sketch Storyboards into 3D Animation.*" ACM Transactions on Graphics (In Proc. of SIGGRAPH). 2025.

[16] Pinyoanuntapong, Ekkasit, Usama Saleem, Korrawe Karunratanakul, Pu Wang, Hong Fei Xue, Chen Chen, **Chuan Guo**, Junli Cao, Jian Ren, Sergey Tulyakov†."*MaskControl: Spatio-Temporal Control for Masked Motion Synthesis*." IEEE International Conference on Computer Vision (ICCV, Best Paper Candidate, 13/11239). 2025. (Accept rate: 24%)

[17] Wang, Yilin, **Guo, Chuan**, Yuxuan Mu, Muhammad Gohar, Xinxin Zuo, Hai Jiang, Juwei Lu, Li Cheng†. "*MotionDreamer: One-to-More Motion Synthesis with Localized Generative Masked Transformer*." In International Conference on Learning Representations (ICLR). 2025. (Accept rate: 32%)

[18] **Guo, Chuan**, Yuxuan Mu, Muhammad Gohar Javed, Sen Wang, Li Cheng†. "*MoMask: Generative Masked Modeling of 3D Human Motions*." In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2024. (Accept rate: 23.6%)

[19] **Guo, Chuan**, Xinxin Zuo, Sen Wang, Xinshuang Liu, Shihao Zou, Minglun Gong, and Li Cheng†. "*Action2video: Generating Videos of Human 3D Actions.*" International Journal of Computer Vision (2022): 1-31.

[20] **Guo, Chuan**, Yuxuan Mu, Xinxin Zuo, Peng Dai, Youliang Yan, Juwei Lu, Li Cheng†. "*Generative Human Motion Stylization in Latent Space*." In International Conference on Learning Representations (ICLR). 2024. (Accept rate: 31%)

[21] Nhat M. Hoang, **Chuan Guo**, Michael Bi Mi and Kehong Gong†. "*MotionMix: Weakly-Supervised Diffusion for Controllable Motion Generation.*" The 38th Annual AAAI Conference on Artificial Intelligence (AAAI). 2024 (Accept rate: 23.75%)

[22] Gong, Kehong, Dongze Lian, Heng Chang, **Chuan Guo**, Xinxin Zuo, Zihang Jang and Xinchao Wang†. "*TM2D: Bimodality Driven 3D Dance Generation via Music-Text Integration*." IEEE International Conference on Computer Vision. 2023. (Accept rate: 26.7%)

[23] **Guo, Chuan**, Xinxin Zuo, Sen Wang, and Li Cheng†. "*TM2T: Stochastic and Tokenized Modeling for the Reciprocal Generation of 3D Human Motions and Texts*." European Conference on Computer Vision (ECCV). 2022. (Accept rate: 28%)

[24] **Guo, Chuan**, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng†. "*Generating Diverse and Natural 3D Human Motion from Text.*" In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). 2022. (Accept rate: 25.3%)

[25] **Guo, Chuan**, Xinxin Zuo, Sen Wang, Shihao Zou, Qingyao Sun, Annan Deng, Minglun Gong, and Li Cheng†. "*Action2Motion: Conditioned Generation of 3D Human Motions.*" In Proceedings of the 28th ACM International Conference on Multimedia, pp. 2021-2029. 2020. (Accept rate: 27.8%)

Topic: Human Pose Estimation

[26] Wang, Li, Yiyu Zhuang, Yanwen Wang, Xun Cao, **Chuan Guo**, Xinxin Zuo, Hao Zhu†. "*Sketch2Posenet: Efficient and Generalized Sketch to 3D Human Pose Prediction.*" In Proceeding of ACM SIGGRAPH Asia.2025.

[27] Zou, Shihao, Xinxin Zuo, Sen Wang, Yiming Qian, **Chuan Guo**, and Li Cheng†. "*Human Pose and Shape Estimation from Single Polarization Images*." IEEE Transactions on Multimedia (2022).

[28] Zou, Shihao, **Chuan Guo**, Xinxin Zuo, Sen Wang, Pengyu Wang, Xiaoqin Hu, Shoushun Chen, Minglun Gong, Li Cheng†. "*EventHPE: Event-based 3-D Human Pose and Shape Estimation*." IEEE International Conference on Computer Vision (ICCV), pp. 10996-11005. 2021. (Accept rate: 25.9%)

Topic: 3D Rendering and Reconstruction

[29] Mir, Aymen, Jian Wang, Riza Alp Guler, **Chuan Guo**, Gerard Pons-Moll, and Bing Zhou†. "*AHA! Animating Human Avatars in Diverse Scenes with Gaussian Splatting*." arXiv preprint arXiv:2511.09827 (2025).

[30] Mu, Yuxuan, Xinxin Zuo, **Guo, Chuan**, Yilin Wang, Juewei Lu, Xiaofei Wu, Songcen Xu, Peng Dai, Youliang Yan, Li Cheng†. "*GSD: View-Guided Gaussian Splatting Diffusion for 3D Reconstruction*." European conference on computer vision (ECCV). 2022. (Accept rate: 27.9%)

Topic: Others (Text mining, saliency detection)

[31] Ji, Wei, Jingjing Li, Qi Bi, **Chuan Guo**, Jie Liu, and Li Cheng†. "*Promoting Saliency From Depth: Deep Unsupervised RGB-D Saliency Detection*." In International Conference on Learning Representations (ICLR). 2022. (Accept rate: 32%)

[32] **Guo, Chuan**, Juan Cao†, Xueyao Zhang, Kai Shu, and Miao Yu. "*Exploiting emotions for fake news detection on social media*." Pre-print (2019).

Honors & Contest

Jul. 2023	**J Gordin Kaplan Graduate Student Award** (1500 CAD)	University of Alberta
Feb. 2023	**Alberta Innovate Graduate Scholarship** (31000 CAD)	Alberta Province
Nov. 2021	**Alberta Graduate Excellence Scholarship** (12000 CAD)	Alberta Province
Oct. 2016	**Qihoo 360 Scholarship** (top 5 out of 1000, 10000 RMB)	Jilin University
Nov. 2015	**National Scholarship** (top 5 out of 281, 8000 RMB)	Ministry of Education
Dec. 2015	Excellent Student of Jilin University	Jilin University
2014, 2015	Second-level Scholarship of Jilin University	Jilin University
2015, 2016	College Excellent Student	Jilin University
May 2015	The 1_{st} Prize in *Jilin Provincial Mathematical Contest in Modeling*	Jilin Province

Academic Talks

Mar. 2026	**Invited Speaker at Simon Fraser University** Topic: Motion Foundation Models and Beyond: Interaction, Physics, and Intelligence	Vancouver, Canada
Mar. 2026	**Invited Speaker at University of Alberta** Topic: Motion Foundation Models and Beyond: Interaction, Physics, and Intelligence	Alberta, Canada
Oct. 2025	**Invited Speaker at ICCV Tutorial** Topic: Mastering Motion Synthesis through Discrete Tokens	Hawaii, USA
Sep. 2025	**Invited Speaker at Tel-Aviv University** Topic: 3D Human Motion Generation from Expressive Texts	Tel Aviv, Israel
Feb. 2025	**Invited Speaker at Northeastern University** Topic: Generative Motion Modeling in Discrete Space	Boston, USA
Feb. 2024	**Invited Speaker at MiHoYo** Topic: 3D Human Motion Generation with Discrete Representation	Montreal, Canada
Dec. 2023	**Invited Speaker at Institute of Computing Technology, CAS** Topic: 3D Human Motion Generation with Discrete Representation	Beijing, China
Dec. 2023	**Invited Speaker at Alberta Machine Intelligence Institute, AI Seminar** Topic: Exploring 3D Human Motions with Deep Learning	Alberta, Canada

Apr. 2023 **Invited Speaker at Computer Vision Meetup** ALBERTA, CANADA

Topic: Generating Diverse and Natural 3D Human Motion from Texts

Dec. 2022 **Invited Speaker at AI TIME Seminar.** CHINA

Topic: Reciprocal Generation of Human Motions and Texts

PROFESSIONAL SERVICE

Organizer

Organizer, ECCV 2025-26 Workshop on Human-informed World Models

Organizer, CVPR 2024-26 Workshop on Human Motion Generation

Organizer, ICCV 2025 Tutorial on 3D Human Motion Generation and Simulation Tutorial

Area Chair

ICML 2026, 3DV 2025, ICLR 2025

Reviewer

Conference Reviewer

ECCV 2024/2026, CVPR 2023-26, SIGGRAPH Asia 2024-25, NeurIPS 2023-25, ICCV 2023,2025, SIGGRAPH 2024-25, ICML 2024-25, Eurographics 2022,2025, ICLR 2024-2025, AAAI 2023-2025, SIGGRAPH Asia 2024-25, MultiMedia 2024, ICCV 2023,2025, ACCV 2022, EMNLP 2021, ACML 2020-2021

Journals Reviewer

IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
IEEE Transactions on Visualization and Computer Graphics (TVCG)
IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
IEEE Robotics and Automation Letters (RA-L)
IEEE Transactions on Multimedia (TMM)
IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
Pattern Recognition (PR)
Machine Learning